I-POWER TORSION MODULES FORM A SERRE SUBCATEGORY

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Allowed standard prerequisites (editorial).

- The extension of two I-power torsion modules is I-power torsion. invoked at proof lines 24079-24081. Provides the third Serre condition.

Source reference. Original definition 02MO: sources/stacks-chow/tags.txt, source lines 3451–3451. Referenced at statement lines 24071-24072 for the notion being verified; the definition is used inside the counted span.

Source reference. Original lemma 05EB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24042–24042. Cited at proof lines 24080-24081 for the closure under extensions; the conclusion is used inside the counted span.

Source reference. Original lemma 02MP: sources/stacks-chow/tags.txt, source lines 3452–3452. Cited at proof lines 24081-24082 to conclude the Serre property; the conclusion is used inside the counted span.

Lemma 0.2. *Let I be a finitely generated ideal of a ring R. The I-power torsion modules form a Serre subcategory of the abelian category Mod_R, see Homology, Definition 02MO.*

Proof. It is clear that a submodule and a quotient module of an I-power torsion module is I-power torsion. Moreover, the extension of two I-power torsion modules is I-power torsion by Lemma 05EB. Hence the statement of the lemma by Homology, Lemma 02MP. □

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